

Electrooculography: Recording and Analysis of Eye Movements Using Surface Electrodes

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Abstract—Electrooculography (EOG) records eye position and movement by detecting the corneoretinal potential, the standing electrical difference between the positively charged cornea and the negatively charged retina, using surface skin electrodes. This study recorded and analyzed EOG signals across six experimental tasks: baseline zeroing, artefact recognition, angular calibration, saccadic recording during reading, smooth tracking, and gaze-holding. Calibration confirmed a strong linear relationship between EOG amplitude and gaze angle across $\pm 45^\circ$ ($R^2 = 0.99$, sensitivity $5.76 \mu\text{V}/^\circ$). Three artefact types were identified: blinking, tight eye closure, and electromyography (EMG) contamination from teeth clenching. A waveform consistent with vestibular nystagmus was observed, characterized by slow compensatory phases and rapid saccadic resets. Although the VOR is physiologically expected to provide superior gaze stability due to its shorter latency, the reading-time results did not clearly reflect this, most likely due to electrode-motion artefacts during head movement. These findings highlight both the utility and the methodological considerations of EOG as a non-invasive tool for eye movement assessment.

Index Terms—Electrooculography, corneoretinal potential, saccades, smooth pursuit, vestibulo-ocular reflex, optokinetic reflex, vestibular nystagmus, signal artefacts, ADInstruments EOG Pod ML317, PowerLab 15T.

I. INTRODUCTION

The human eye is never truly still. Even during what feels like a steady gaze, six extrinsic muscles per eye are constantly working to keep the image focused on the fovea, the small central region of the retina responsible for the sharpest vision. These movements are organized into five functional subsystems: saccades, smooth pursuit, the vestibulo-ocular reflex (VOR), the optokinetic reflex (OKR), and vergence, each serving a distinct role in maintaining visual clarity as the head and visual environment move [1], [2].

What makes this system particularly interesting from a biomedical engineering perspective is that every rotation of the eye generates a measurable electrical signal at the surface of the face. The cornea carries a positive charge relative to the retina, producing a steady standing potential of approximately 0.4 to 1 mV known as the corneoretinal potential [3]. As the eye rotates, this electrical dipole shifts with it, and skin electrodes placed around the eye detect the resulting voltage change. This is the principle behind

electrooculography (EOG), formally described by Marg in 1951 [4] and now a standard tool for studying eye movements and assessing retinal pigment epithelium (RPE) function [1], [5]. The ISCEV maintains a clinical EOG standard specifying electrode placement, impedance targets, and amplifier requirements [6].

EOG requires only surface electrodes and a signal amplifier, making it more accessible than video-oculography (VOG) [7], and has been adapted for assistive devices enabling motor-impaired users to control computers through eye movement [8]. The trade-off is susceptibility to artefacts from blinks, jaw clenching, and head movement, and recognizing these patterns is a key skill in biomedical signal acquisition [9]. Six tasks were completed: (1) baseline zeroing; (2) artefact identification; (3) angular calibration; (4) saccadic recording during reading; (5) smooth pursuit; and (6) VOR and OKR gaze-holding.

II. MATERIALS AND METHODS

The experimental setup is shown in Fig. 1. Tasks were ordered progressively from baseline establishment to increasingly complex conditions.

A. Participant

One healthy adult male volunteer (age approximately 23 years, right-handed, normal uncorrected vision) with no known history of ocular, neurological, or vestibular pathology participated. Verbal consent was obtained in accordance with Keele University practical session guidelines.

B. Equipment and Electrode Setup

EOG signals were recorded using an ADInstruments EOG Pod (ML317; ADInstruments Pty Ltd, Sydney, Australia) connected to a PowerLab 15T data acquisition unit (PL15T02; ADInstruments Pty Ltd, Sydney, Australia) interfaced with a laptop running the Lt platform [10]–[12]. The EOG Pod used DC coupling with a gain of $\times 1000$ and a 500 Hz low-pass filter, while the PowerLab incorporated isolated input circuitry compliant with IEC 60601-1 requirements for body-applied parts [11]. Skin preparation followed ISCEV impedance guidelines [6], although impedance was not quantitatively verified.

Three skin sites were prepared: the forehead (ground) and the outer canthus of each eye. The skin was abraded with Nuprep ECG & EEG Abrasive Skin Prepping Gel (Weaver and Company, Aurora, CO, USA) and cleaned using Briemarpak Skin Cleansing Swabs (Briemarpak Ltd, UK) to minimize electrode impedance in accordance with ISCEV-recommended $25 \text{ k}\Omega$ [6]. Disposable Ag/AgCl electrodes were applied because of their low noise and electrochemical stability [6]. The positive (white) electrode

was placed at the right outer canthus, the negative (black) electrode at the left outer canthus, and the ground (green) electrode at the forehead midline, forming a horizontal canthal derivation that recorded changes in the corneoretinal potential during horizontal eye movement [1], [6].

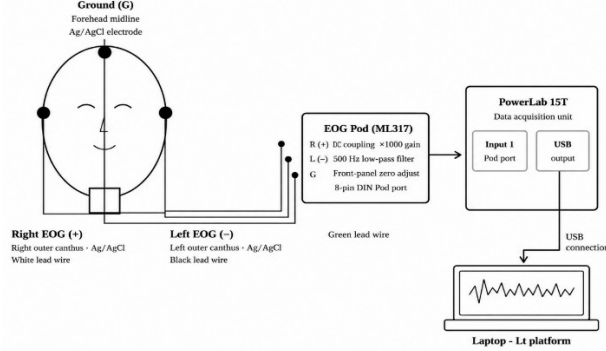


Fig. 1. Schematic of the EOG experimental setup. Ag/AgCl electrodes at the right outer canthus (positive), left outer canthus (negative), and forehead midline (ground) connect via lead wires to the EOG Pod (ML317). The amplified signal feeds to the PowerLab 15T via the Pod port, which connects via USB to a laptop running the Lt platform.

C. Experimental Procedures

The EOG signal was re-zeroed before each task to correct for DC drift [12]. Tasks were ordered deliberately: zeroing and artefact recognition first to confirm a clean signal, then calibration to convert voltages into angular units, followed by saccadic, tracking, and gaze-holding tasks. Zeroing was performed with the volunteer gazing straight ahead until the output reached approximately 0 mV. Three artefact conditions were then recorded and annotated: normal blinking, sustained eye closure (5 s), and teeth clenching (5 s). For calibration, the stool was positioned 1 m from the table edge and targets at $\pm 45^\circ$, $\pm 30^\circ$, $\pm 15^\circ$, and 0° were marked on the table surface; the volunteer fixated each in turn while EOG amplitude was recorded. The volunteer then read a text passage for saccadic recording, tracked a pen at constant velocity for smooth pursuit, and performed OKR (hand moving, head still) then VOR (head rotating, hand still) at increasing speed; paragraph reading time was measured under both conditions.

III. RESULTS

The results follow the same order as the tasks: baseline zeroing confirmed signal stability; calibration converted voltages into gaze angles; and the subsequent recordings were interpretable because those foundations were established first.

A. Baseline and Artefact Recognition

The zeroing procedure settled the baseline at 0.0271 mV. Three artefact conditions were then recorded, each producing a distinctly different signal. Blinking caused short, biphasic deflections; tight eye closure (5 s) produced a sustained high-amplitude disturbance; and teeth

clenching generated irregular, higher-frequency bursts reaching approximately -0.354 mV. With these signatures identified, similar patterns in later recordings could be confidently attributed to artefacts rather than genuine eye movements.

B. Angular Displacement Calibration

Calibration converted raw EOG voltage into gaze angle. Amplitude increased progressively with rightward gaze and decreased with leftward gaze (Table I), and linear regression yielded $R^2 = 0.99$ with a slope of $5.76 \mu\text{V}/^\circ$, demonstrating strong linearity across the seven fixation angles. A slight compression at $\pm 45^\circ$ (Fig. 2) reflected the physical limits of lateral eye movement. All subsequent deflections could now be expressed in angular terms, making the recordings that followed directly interpretable.

TABLE I. EOG Amplitude at Known Angular Displacements

Angular Displacement ($^\circ$)	EOG Amplitude (mV)
-45	-0.198
-30	-0.132
-15	-0.0389
0 (central gaze)	0.0104
+15	0.123
+30	0.238
+45	0.308

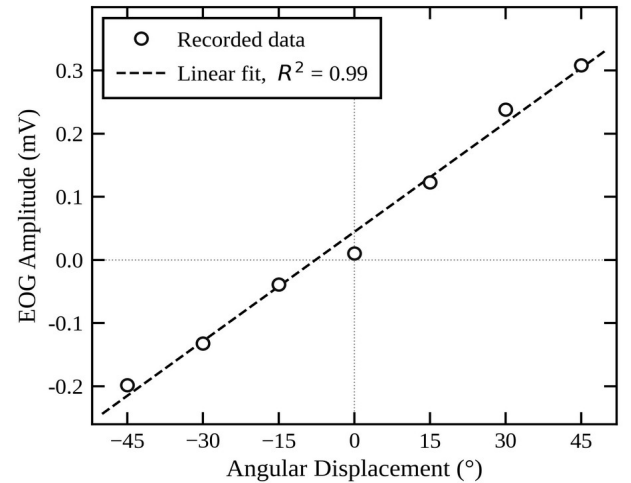


Fig. 2. Angular displacement vs EOG amplitude. Open circles are recorded data; dashed line is the linear fit. Slope = $5.76 \mu\text{V}/^\circ$, $R^2 = 0.99$.

C. Saccadic Eye Movements During Reading

With a calibrated signal in place, the reading task revealed how the eyes move through text. The recorded EOG signal demonstrated repeated rapid deflections corresponding to sequential forward eye movements as the volunteer read each line. Five distinct reading intervals were identified (Table II, mean 4.26 s, SD 1.17 s), each representing a

sequence of fixations and forward saccades. The largest deflection corresponded to the end-of-line return sweep.

TABLE II. Reading Interval Durations During Saccadic Recording (Mean $\pm$ SD = 4.26 $\pm$ 1.17 s)

Interval No.	Duration (s)
1	5.78
2	4.55
3	4.51
4	3.94
5	2.54
Mean	4.26
SD	1.17

D. Smooth Tracking

In contrast to the abrupt deflections seen during reading, the recorded EOG signal during smooth tracking demonstrated a gradual, continuous amplitude change as the pen moved leftward, shifting from approximately +0.1 mV toward negative values relative to baseline. Brief superimposed deflections consistent with low-amplitude corrective saccades were observed intermittently during target pursuit, indicating transient retinal slip correction as the eye momentarily lagged behind the moving target.

E. Gaze-Holding: VOR and OKR

The gaze-holding tasks assessed the ability of the oculomotor system to maintain a stable image during head and visual scene movement. During Procedure 1, the recorded EOG signal peaked at 0.477 mV, and a sawtooth waveform consistent with vestibular nystagmus was observed, characterized by slow compensatory phases alternating with rapid saccadic resets. During Procedure 2, reading time was 0.31 s with head shaking (VOR) and 0.27 s with the head still and book moving (OKR) (Table III). These values likely reflect a short phrase rather than a full paragraph, consistent with the Lt platform task format.

TABLE III. Reading Time Under Two Gaze-Holding Conditions

Condition	Reading Time (s)
Head moves, book still (VOR)	0.31
Head still, book moves (OKR)	0.27

IV. DISCUSSION

Each finding is discussed against established oculomotor physiology, noting where results diverge.

A. Calibration and Signal Linearity

The near-perfect linearity ($R^2 = 0.99$) across $\pm 45^\circ$ is consistent with the corneoretinal dipole model, which

predicts a proportional relationship between gaze angle and EOG amplitude within the central range of eye movement [1], [3]. The recorded sensitivity of 5.76 $\mu\text{V}/^\circ$ is lower than the 20 $\mu\text{V}/^\circ$ reported for clinical EOG configurations [6], [9]. The compact canthal placement captures a smaller fraction of the dipole field than wider periorbital setups, reducing recorded voltage per degree; sensitivity values cannot therefore be compared across studies without knowing electrode configuration. Calibration residuals remained small across the central $\pm 30^\circ$ range, supporting the assumption of linearity within physiologically typical gaze angles. The residual of 0.0104 mV at central gaze reflects DC drift that re-zeroing alone does not fully eliminate [6].

B. Artefacts: Origins and Clinical Relevance

Each artefact has a distinct physiological origin. Blink artefacts arise from Bell's phenomenon: eyelid closure rotates the globe upward, producing a deflection mistakable for a real gaze shift [9]; Kirchner et al. confirmed that blinking tends to follow rather than interrupt saccades [20]. Teeth clenching generated the most striking signal: masticatory EMG travels through the facial tissue to the periocular electrodes, appearing as high-frequency bursts [9]. Clinically, this could be mistaken for RPE dysfunction, yet the same signal is deliberately exploited as a control command in EOG-based assistive devices [8]. Head movement was the most disruptive: electrode-skin displacement introduces transients that can obscure the signal entirely [7], requiring chin rest immobilization in clinical practice.

C. Saccades During Reading

The 4.26 s mean reading interval does not reflect individual saccade duration. Saccades last 20 to 200 ms, reaching velocities up to 700 $^\circ$ /s [13], [14], and are invisible because vision is suppressed during each movement via saccadic suppression [2], [15]. The intervals in Table II instead capture the time between major gaze-shift events, each spanning one or more fixation periods plus multiple small forward saccades. The end-of-line return sweep, which produced the largest single deflection, is a recognized marker of reading strategy and line-length adaptation [16].

D. Smooth Tracking and Corrective Saccades

The gradual, continuous EOG decline during tracking reflects smooth pursuit, which minimizes retinal slip by matching eye velocity to target motion [17]. The small corrective deflections superimposed on this trace are not noise; they are normal catch-up saccades, typically 0.5 $^\circ$ to 2 $^\circ$, that the oculomotor system generates automatically when the eye falls behind the target [17], [15]. Observing both in the same trace illustrates how the oculomotor system integrates separate subsystems to maintain continuous foveal alignment.

E. Gaze-Holding: VOR, OKR, and Nystagmus

The sawtooth waveform was consistent with vestibular nystagmus driven by the VOR: slow compensatory eye movements counteracting head rotation, interrupted by

rapid saccadic resets each time the eye reached its orbital limit [18]. The VOR operates via a direct three-neuron brainstem arc (~10 ms latency), bypassing cortical processing, while the OKR depends on visual cortex with a latency of 80 to 100 ms [19]. The VOR should therefore provide more stable gaze during rapid movement. The reading-time results (0.31 s for VOR versus 0.27 s for OKR) do not reflect this expectation, which is most plausibly explained by electrode-motion artefacts introduced by head movement rather than true physiology. The 40 ms difference is too small to be meaningful given the single-volunteer design, and these results do not contradict established oculomotor physiology.

F. Limitations

Only one volunteer was tested, preventing statistical generalization. The ISCEV 2017 standard [6] and IEC 60601-1 electrical safety requirements [11] formed the governance framework for this recording; however, the absence of formal ethics approval and quantitative impedance verification represent departures from clinical EOG deployment standards. Ambient lighting was uncontrolled despite the ISCEV recommendation of at least 200 lux to ensure corneoretinal potential stability [6]. The EOG Pod's 500 Hz filter limits temporal resolution for saccadic peak-velocity measurement compared with dedicated high-bandwidth systems [21], and only horizontal eye movements were recorded; vertical movements, which carry additional clinical information such as Bell's phenomenon quantification, were not captured [1], [6]. Dark adaptation was also not controlled; the corneoretinal potential is sensitive to illumination history, a factor central to the clinical Arden ratio used to assess RPE function [6]. A more rigorous design would standardize lighting, deploy two EOG Pods for simultaneous horizontal and vertical recording, and use a chin rest.

V. CONCLUSIONS

Surface electrode EOG reliably distinguished saccades, smooth pursuit, and gaze-holding reflexes within a single session. Calibration confirmed $R^2 = 0.99$ ($5.76 \mu\text{V}^\circ$), consistent with the ISCEV standard [6]; the lower sensitivity reflects the compact canthal electrode spacing. All three artefact types were identifiable, each with a physiological origin relevant to clinical EOG interpretation. Vestibular nystagmus was directly observed as a sawtooth waveform, confirming VOR-driven gaze compensation. The reading-time results did not demonstrate the VOR's expected latency advantage, attributed to electrode-motion artefacts rather than genuine physiology [18], [19]. These findings reinforce a core principle: meaningful physiological measurement requires a stable signal, rigorous calibration, and the ability to separate real physiology from noise.

REFERENCES

- [1] D. J. Creel, "The electrooculogram," *Handb. Clin. Neurol.*, vol. 160, pp. 495–499, 2019, doi: 10.1016/B978-0-444-64032-1.00033-3.
- [2] D. C. Niehorster et al., "Vision toolkit part 1: neurophysiological foundations and experimental paradigms in eye-tracking research," *Front. Hum. Neurosci.*, vol. 19, p. 1547677, 2025, doi: 10.3389/fnhum.2025.1547677.
- [3] A. Bulling, J. A. Ward, H. Gellersen, and G. Tröster, "Eye movement analysis for activity recognition using electrooculography," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 4, pp. 741–753, Apr. 2011, doi: 10.1109/TPAMI.2010.86.
- [4] E. Marg, "Development of electrooculography: standing potential of the eye in registration of eye movement," *AMA Arch. Ophthalmol.*, vol. 45, pp. 169–185, 1951.
- [5] H. D. Katzberg and A. Abraham, "Electrooculography," in *Encyclopedia of Sleep and Circadian Rhythms*, 3rd ed. Amsterdam: Elsevier, 2023, pp. 474–477.
- [6] P. A. Constable, M. Bach, L. J. Frishman, B. G. Jeffrey, and A. G. Robson, "ISCEV standard for clinical electrooculography (2017 update)," *Doc. Ophthalmol.*, vol. 134, no. 1, pp. 1–9, Feb. 2017, doi: 10.1007/s10633-017-9573-2.
- [7] J. Colebatch, H. MacDougall, and M. Halmagyi, "Comparison of EOG and VOG obtained eye movements during horizontal head impulse testing," *Front. Neurol.*, vol. 13, p. 917413, Aug. 2022, doi: 10.3389/fneur.2022.917413.
- [8] W.-D. Chang, "Electrooculograms for human-computer interaction: a review," *Sensors*, vol. 19, no. 12, p. 2690, Jun. 2019, doi: 10.3390/s19122690.
- [9] S. S. Plöchl, J. P. Ossandón, and P. König, "Combining EEG and eye tracking: identification, characterisation, and correction of eye movement artefacts in electroencephalographic data," *Front. Hum. Neurosci.*, vol. 6, p. 278, Oct. 2012, doi: 10.3389/fnhum.2012.00278.
- [10] ADInstruments, "EOG Pod ML317 data card," ADInstruments Pty Ltd, Sydney, Australia. [Online]. Available: <https://m-cdn.adinstruments.com/product-data-cards/ML317-DCW-15A.pdf>. [Accessed: 31 Mar. 2026].
- [11] ADInstruments, "PowerLab 15T (Model PL15T02) education PowerLab," ADInstruments Pty Ltd. [Online]. Available: <https://www.adinstruments.com/products/powerlab/education>. [Accessed: 31 Mar. 2026].
- [12] ADInstruments / Lt Platform, "Electrooculography (EOG) – Lab," MTE-40026 Physiological Measurements Practicals, ADInstruments Pty Ltd, kuracloud.com. Lesson committed 26 Mar. 2026.
- [13] A. T. Bahill, M. R. Clark, and L. Stark, "The main sequence, a tool for studying human eye movements," *Math. Biosci.*, vol. 24, no. 3–4, pp. 191–204, 1975, doi: 10.1016/0025-5564(75)90075-9.
- [14] Z. Nouri and N. Mikaeili, "Electrooculography-based saccadic eye movement detection using adaptive thresholding and peak analysis," *J. Med. Signals Sens.*, vol. 12, no. 1, pp. 8–16, Jan. 2022, doi: 10.4103/jmss.jmss_30_20.
- [15] S. Oh and J.-H. Kim, "Recording and interpretation of ocular movements: saccades, smooth pursuit, and optokinetic nystagmus," *Ann. Clin. Neurophysiol.*, vol. 25, no. 2, pp. 55–65, Oct. 2023, doi: 10.14253/acn.2023.25.2.55.
- [16] J. Chandra, A. Krügel, and R. Engbert, "Modulation of oculomotor control during reading of mirrored and inverted texts," *Sci. Rep.*, vol. 10, p. 4210, Mar. 2020, doi: 10.1038/s41598-020-60833-6.

- [17] A. Goettker and K. R. Gegenfurtner, "A change in perspective: the interaction of saccadic and pursuit eye movements in oculomotor control and perception," *Vision Res.*, vol. 188, pp. 283–296, Nov. 2021, doi: 10.1016/j.visres.2021.08.004.
- [18] D. Yüksel, B. Gümüşöz, and B. Akdal, "Vestibulo-ocular reflex: a narrative review," *Matrix Sci. Med.*, vol. 7, no. 4, pp. 90–94, Oct.–Dec. 2023, doi: 10.4103/mtsm.mtsm_24_22.
- [19] T. Mergner, G. Schweigart, S. Morand, I. Evdokimidis, and F. Hlavacka, "Interaction of optokinetic reflex and vestibulo-ocular reflex during active and passive head rotation," in *Multisensory Control of Posture*. New York: Plenum, 1995, pp. 201–210.
- [20] J. Kirchner, T. Watson, N. A. Busch, and M. Lappe, "Timing and kinematics of horizontal within-blink saccades measured by EOG," *J. Neurophysiol.*, vol. 127, no. 6, pp. 1655–1668, Jun. 2022, doi: 10.1152/jn.00076.2022.
- [21] R. Wierds, M. J. Janssen, and H. Kingma, "Measuring saccade peak velocity using a low-frequency sampling rate of 50 Hz," *IEEE Trans. Biomed. Eng.*, vol. 55, no. 12, pp. 2840–2842, Dec. 2008, doi: 10.1109/TBME.2008.2001284.